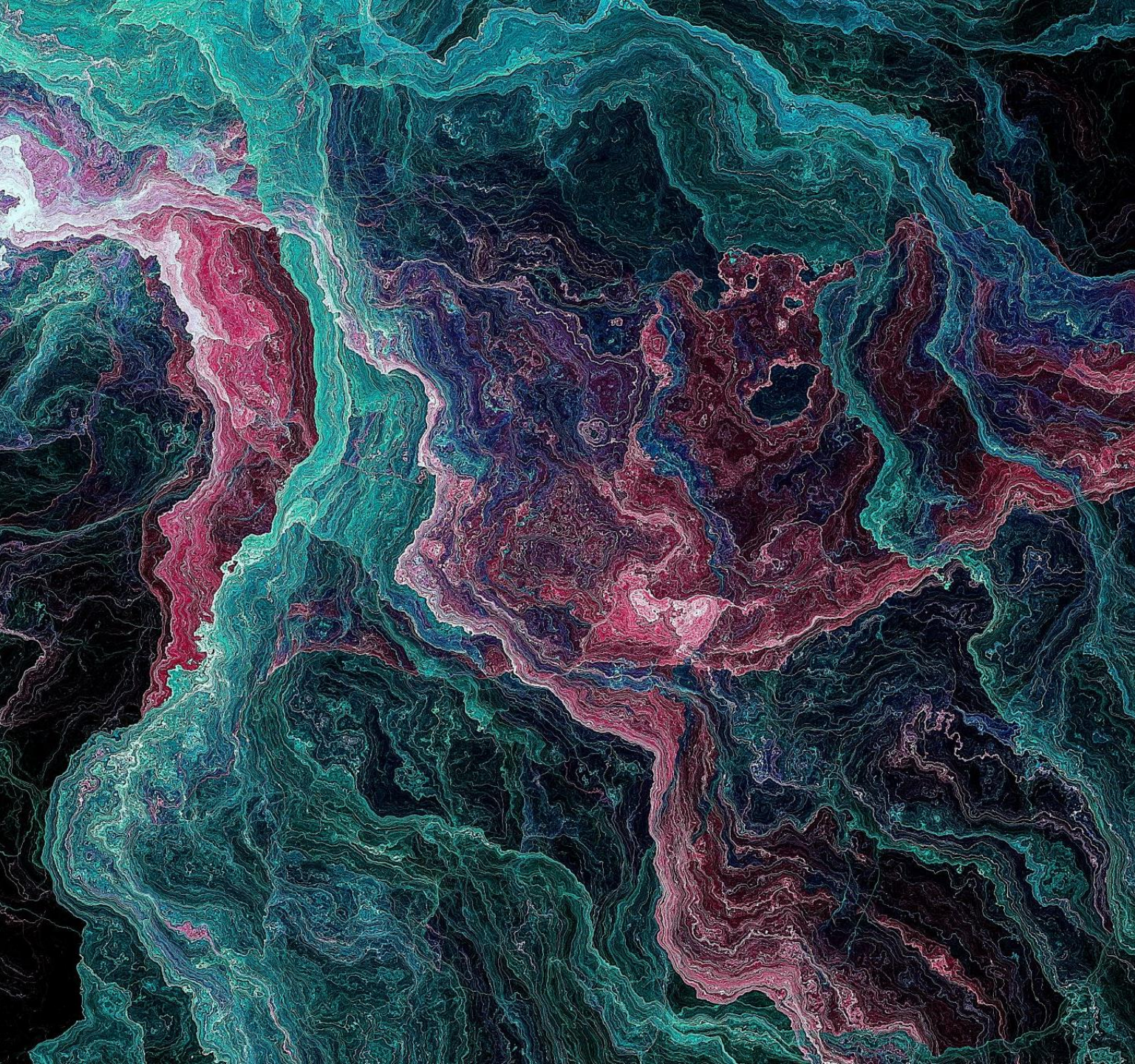




Tim Houle, PhD

ADVANCED ISSUES IN CLINICAL TRIALS

RETRACTION RECORD ROCKS COMMUNITY

ANESTHESIOLOGY TRIES TO MOVE ON AFTER FRAUD INVESTIGATIONS



Breathe deeply: anaesthetists are well placed to conduct clinical trials with relatively little oversight.

FRAUD IN ANESTHESIOLOGY

- Scott Reuben, MD (1996 to 2010)
 - 21 fraudulent scientific articles retracted
 - Admits that he never conducted any of the trials he reported in the literature
 - Sentenced to prison (2010)
 - Joachim Boldt, MD (1999 to 2012)
 - False data in 88 (of 102) scientific articles retracted
 - Yoshitaka Fujii, MD (1993 to 2012)
 - 172 (of 212) fraudulent scientific articles, with 126 articles entirely fabricated
 - No ethical oversight, authorship inclusions
-

DETECTING FABRICATED DATA IN RCTS

- J.B. Carlisle (2012)* applied a clever strategy to examine the probability that Fujii's data were the result of non-random processes
 - Distribution of patient characteristics “incredibly nice” (Kranke et al, 2000)
 - Patients in randomized groups *too similar* to each other
- Examined 168 of Fujii RCTs
 - Compared expected distributions to observed distributions of patient characteristics

*Carlisle (2012). The Analysis of 168 Randomised Controlled Trials to Test Data Integrity

THE “TABLE 1” OF AN RCT

Table 1. Patient and Labor Characteristics of Laboring Patients

	Study group (neostigmine + bupivacaine) (<i>n</i> = 20)	Control group (bupivacaine alone) (<i>n</i> = 20)
Age (yr)	27 ± 6	29 ± 7
Height (cm)	166 ± 5	164 ± 9
Weight (kg)	84 ± 14	84 ± 15
Gestation (wk)	39 ± 1	39 ± 1
Parous (%)	45%	45%
Oxytocin use (%)	55%	75%
Oxytocin infusion rate (mU/min) at time of epidural catheter placement ^a	7.0 ± 10.3	8.4 ± 7.6
Fetal weight (g)	3350 ± 540	3370 ± 450
VRS score-baseline ^b	7.6 ± 1.8	6.9 ± 2.2
VRS score-0 min ^c		
Mean ± SD	0.6 ± 1.3	0.8 ± 1.3
Median	0	0
Mode	0	0
Range	0–3	0–3
Cervical dilation prior epidural placement (cm)	3.1 ± 1.3	3.3 ± 1.3
Cervical dilation 2 h after epidural placement (cm)	4.7 ± 1.5	4.3 ± 1.5

Data are presented as mean ± SD or incidence (%) unless otherwise indicated. There are no statistical differences between groups for any variable.

VRS = verbal rating scale.

^a Average of all patients including those with or without oxytocin use at time of epidural catheter placement.

^b VRS before initiation of epidural analgesia.

^c Twenty minute after injection of epidural study solution.

THE “TABLE 1” OF AN RCT

The standardized difference between groups should be normally distributed with A mean of 0.

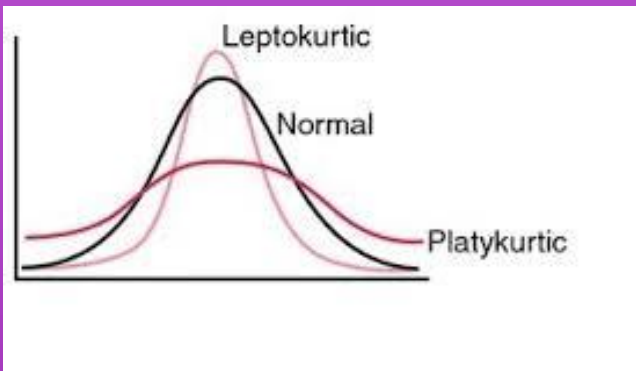


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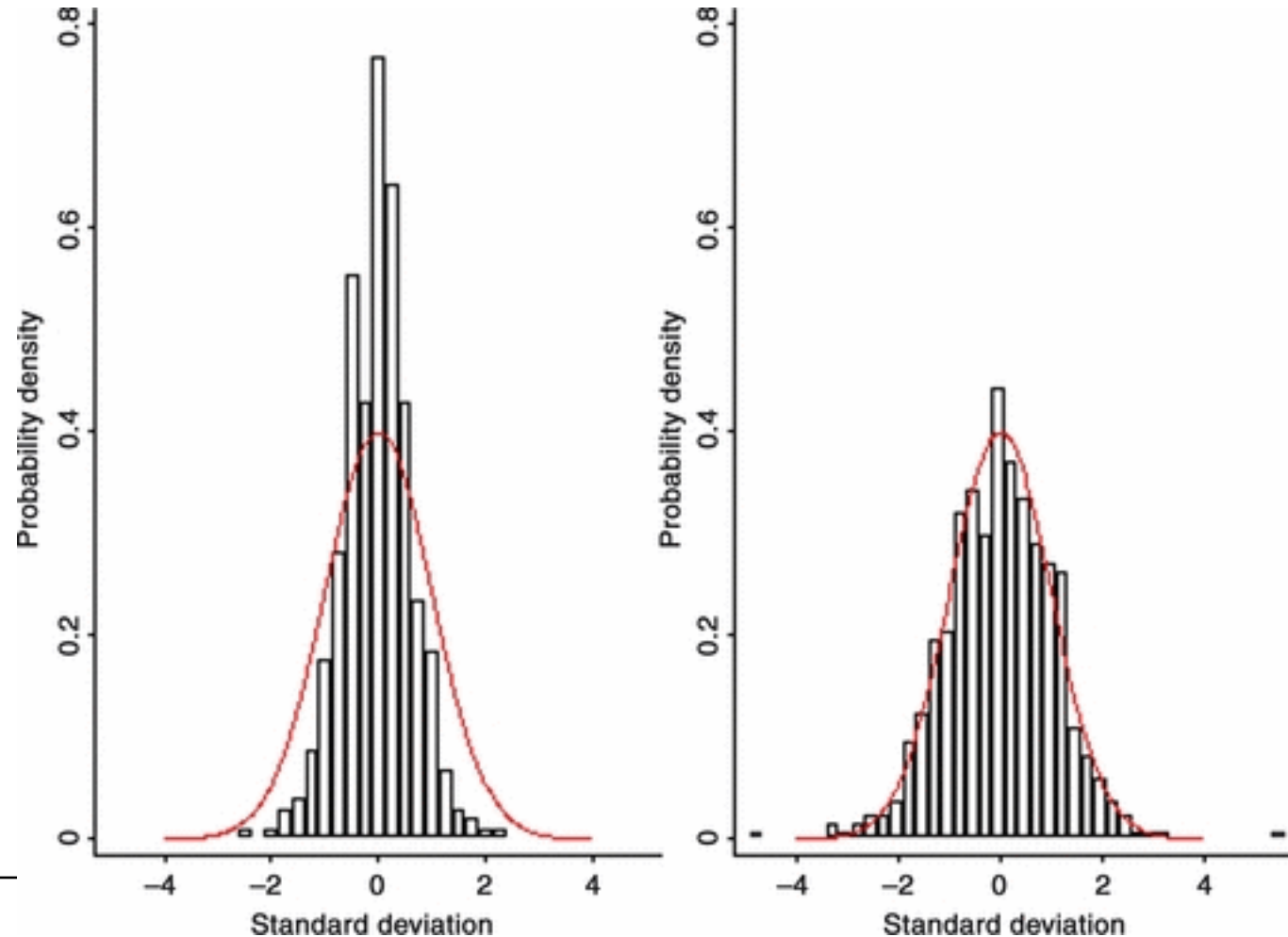
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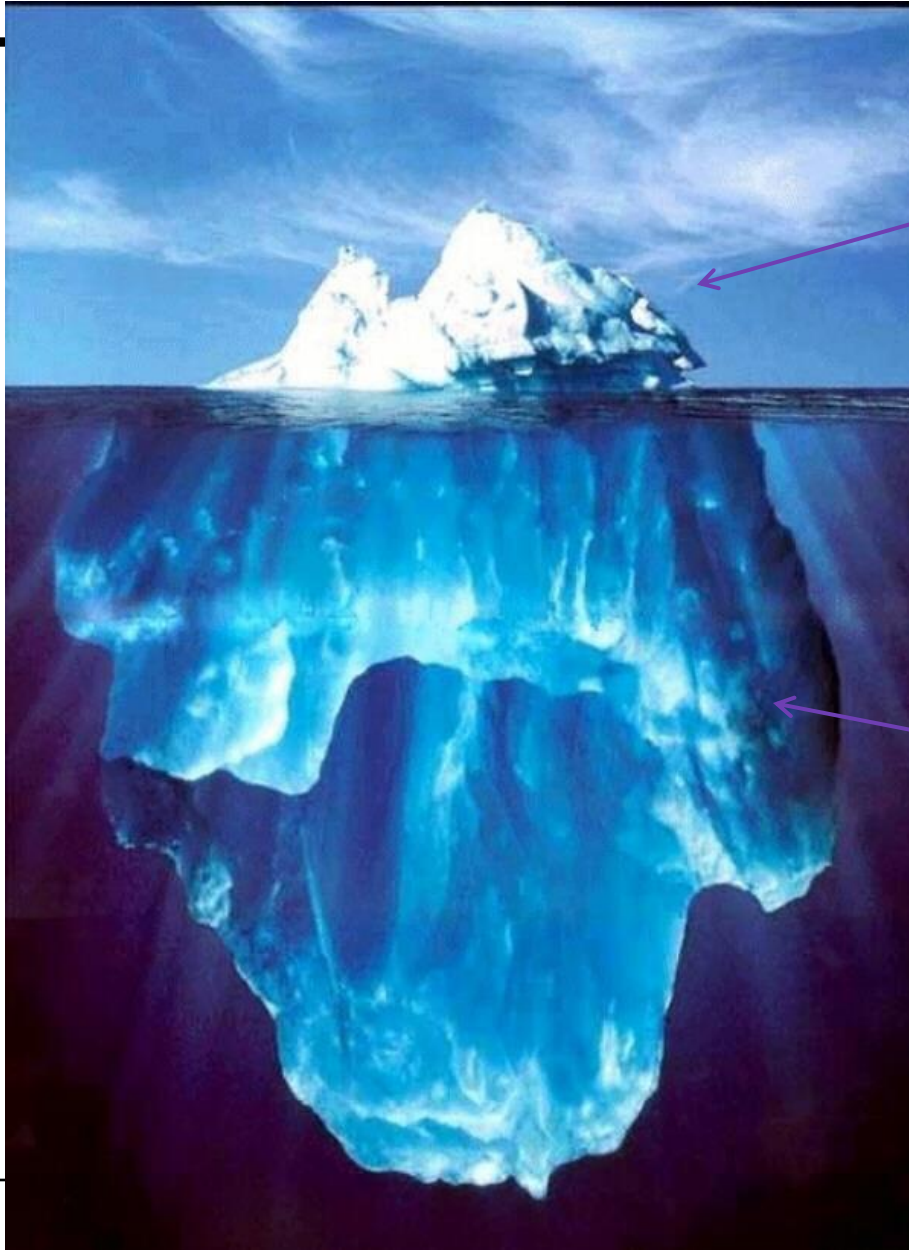
^a Average of all patients including those with or without oxytocin use at time of epidural catheter placement.

^b VRS before initiation of epidural analgesia.

^c Twenty minute after injection of epidural study solution.

FUJII (LEFT) VERSUS OTHERS (RIGHT) ON AGE





Data Fabrication/Falsification

Poor research conduct
(Conscious or Unconscious)

POOR QUALITY OF MEDICAL RESEARCH

- J Cohen (1962): Statistical power for medium effects in social science literature = 48%
 - D Altman (2001):
 - Failure to specify eligibility criteria in 25% of surgery Journals
 - Failure to disclose the nature of blinding in 33%- 51% of RA journals, cystic fibrosis journals, and dermatology
 - Inadequate information on harmful consequences of interventions in 61% of Journals
-

LOW QUALITY MEDICAL RESEARCH: RANDOMIZED CONTROL TRIALS

- 1: de Vries TW, van Roon EN. Low quality of reporting adverse drug reactions in paediatric randomised controlled trials. Arch Dis Child. 2010 Jun 15.
 - 2: Zhang D, Freemantle N, Cheng KK. Are randomized trials conducted in China or India biased? A comparative empirical analysis. J Clin Epidemiol. 2010 Jun 14.
 - 3: Pandis N, Polychronopoulou A, Eliades T. An assessment of quality characteristics of randomized control trials published in dental journals. J Dent. 2010 May 20.
 - 4: Delaney M, Meyer E, Cserti-Gazdewich C, Haspel RL, Lin Y, Morris A, Pavenski K, Dzik WH, Murphy M, Slichter S, Wang G, Dumont LJ, Heddle N. A systematic assessment of the quality of reporting for platelet transfusion studies. Transfusion. 2010 May 24.
 - 5: Yu LM, Chan AW, Hopewell S, Deeks JJ, Altman DG. Reporting on covariate adjustment in randomised controlled trials before and after revision of the 2001 CONSORT statement: a literature review. Trials. 2010 May 18;11:59.
 - 6: Friction JR, Ouyang W, Nixdorf DR, Schiffman EL, Velly AM, Look JO. Critical appraisal of methods used in randomized controlled trials of treatments for temporomandibular disorders. J Orofac Pain. 2010 Spring;24(2):139-51.
-

Why Most Published Research Findings Are False -Ioannidis (2005)

SOURCES OF POOR QUALITY RESEARCH REPORTS: REPORTING ISSUES

- Inadequate or incomplete reporting
 - Example: Failure to disclose the unblinded nature of a trial
- Errors of fact or interpretation
 - Example: using wrong analysis
- Bias
 - Example: Spin

Inadequate or poor research design issues
are paramount!

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KEY RESOURCES

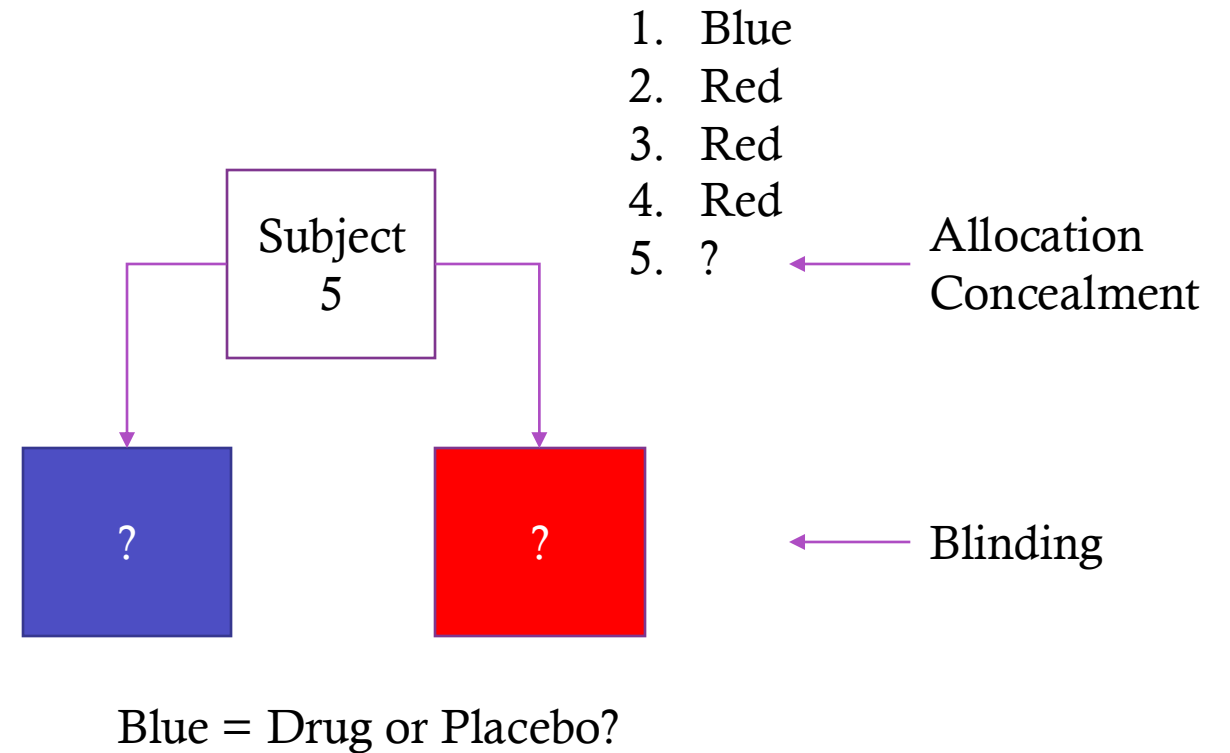
- SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials)
 - <https://www.spirit-statement.org/>
 - CONSORT (Consolidated Standards of Reporting Trials)
 - <http://www.consort-statement.org/>
 - CONSORT – SPI (2018) for Social and psychological interventions
 - Guidelines for the Content of Statistical Analysis Plans in Clinical Trials
 - <https://jamanetwork.com/journals/jama/fullarticle/2666509>
-

SEVERAL EXAMPLES

- Randomization & Concealment
 - Statistical Power
 - Analysis
 - ‘Spin’
-

BLINDING VS ALLOCATION CONCEALMENT

- Not the same thing!



TYPES OF ALLOCATION CONCEALMENT

- Centralized allocation assignments
 - Investigators call the pharmacy with a subject number and get a treatment
 - Pre-programmed web app that requires 'breaking the seal' to get assignment
 - Sealed, opaque, envelopes
 - Assignment written inside on a sheet of paper
 - Just in time randomization
 - Flip a coin (with observers) to make an assignment at the required time
-

TYPES OF BLINDING (MASKING)

- Unblinded (Open-label)
 - Single-blinded
 - Only participant is unaware of the assignment
 - Double-blind
 - Participant and investigator(s)
 - Triple Blind
 - Participant, investigator, adjudicator
-

TYPES OF RANDOMIZATION

- Simple
 - Without restrictions
 - Restricted
 - Blocked randomization (e.g., blocks of 4)
 - Stratified randomization (e.g., males, females)
 - Dynamic or adaptive
 - Minimization
-

CONSORT REPORTING

(8a) Methods used: to generate the randomization sequences (e.g., computer generated list of random numbers)

(8b) Type of randomization: report if the randomization assignments were generated without restriction (e.g., simple randomization) or if restrictions were used (e.g., restricted randomization, blocked randomization, stratified randomization, or minimization).

(9) Allocation concealment: report how the foreknowledge of future assignments was concealed from the investigators who were enrolling participants

- How the randomization schedule was concealed from the investigators (e.g., third-party allocation, central randomization, pharmacy administered).
- How the sequence of assignments was administered (e.g., sequentially numbered opaque envelopes).

(10) Randomization Implementation:

- Who generated the allocation sequence (e.g., study statistician)?
 - Who assigned participants to study groups (e.g., the research nurse opened the sequentially numbered envelope containing the randomization assignment)
 - Who enrolled participants into the study (e.g., the research assistant evaluated eligibility, obtained informed consent, and enrolled the participants).
-

WHAT IS STATISTICAL POWER?

- “...The probability of finding significant results... when an effect actually exists”
- Implies:
 - A certain size of effect
 - A certain decision error rate
 - A certain sample size

Compare



NULL HYPOTHESIS SIGNIFICANCE TESTING AND DECISION MAKING

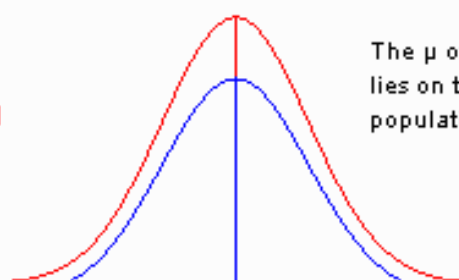
- When conducting a statistical test, an investigator is attempting to *decide* if an effect exists
 - Null hypothesis (H_0): No effect exists
 - Alternative hypothesis (H_1): The effect exists
- Given the consequences of making this decision, what sample size (study design) is needed to produce reliable results?

IQ Test Scores

Null Hypothesis

$$H_0: \text{VA } \mu = \text{US } \mu$$

The μ of the VA distribution lies on the μ of the US population distribution



Alternate Hypothesis

$$H_1: \text{VA } \mu \neq \text{US } \mu$$

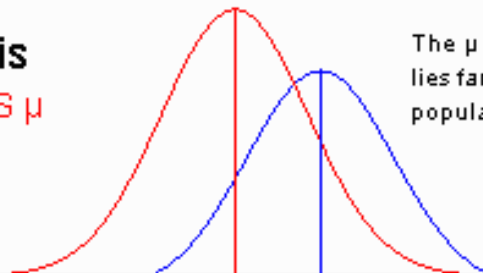
The μ of the VA distribution lies far below the μ of the US population distribution



Alternate Hypothesis

$$H_1: \text{VA } \mu \neq \text{US } \mu$$

The μ of the VA distribution lies far above the μ of the US population distribution



HYPOTHESIS TESTING

$\alpha \rightarrow$ probability of rejecting H_0 when it is true

$\beta \rightarrow$ probability of failing to reject H_0 when it is false

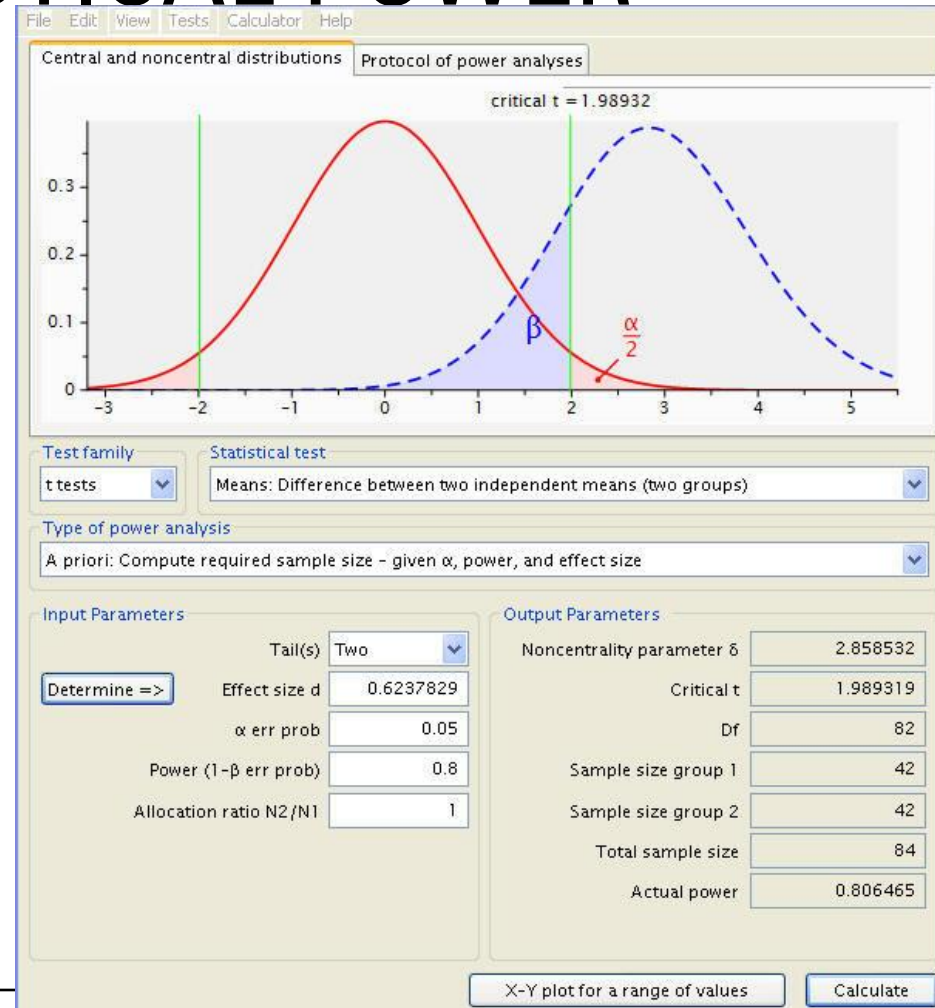
		H_0	
		True	False
Study	Fail to Reject H_0	Correct	β
	Reject H_0	α	Correct

$\beta = \text{Type II error}$
 $1 - \beta = \text{power}$

$\alpha = \text{Type I error}$
 $= \text{corresponds to the p-value}$

RUDIMENTS OF STATISTICAL POWER

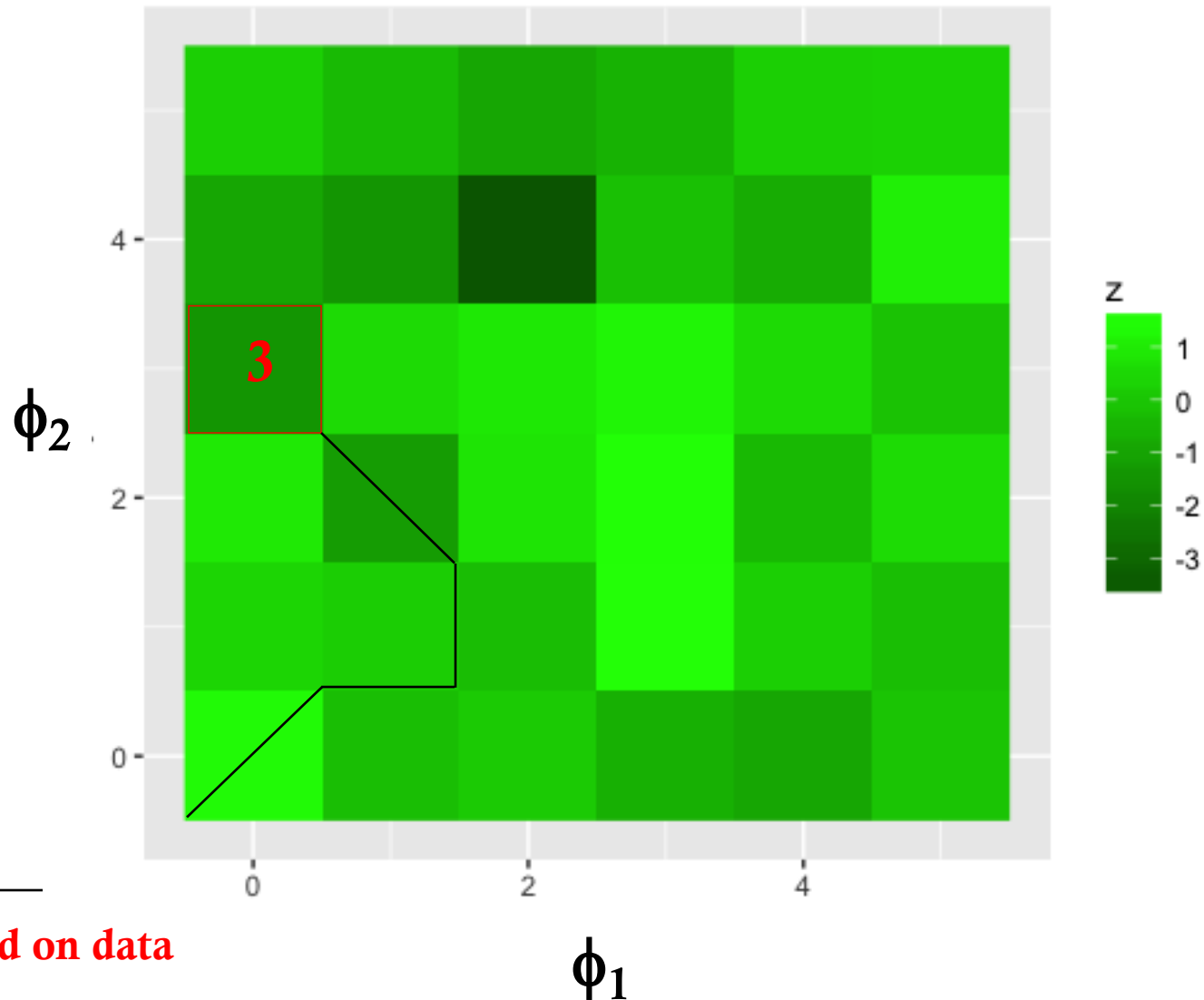
- Effect Size (d)
- Alpha level (α)
- Beta level (B)
- Tailed testing (two)
- Sample size (N)
- Test



CONSORT REPORTING

- (7a) To properly interpret the statistical inferences, a reader needs to be able to evaluate the assumptions used in the statistical power calculation. Please report:
 - the estimated power level (e.g., power = 0.80)
 - the expected effect size (e.g., odds ratio, difference in proportions, mean differences, d, etc.)
 - the variability around the effect using an appropriate estimate of variance (e.g., mean (SD))
 - the required sample size to obtain the desired power (e.g., n = 50 per group)
 - the significance level and nature of hypothesis (e.g., two-sided alpha = 0.05)
-

GARDEN OF FORKING PATHS



Test Based on data

Garden of Forking Paths
Gelman (2013)

	Scenario	Test Statistic	
	1. Simple classical test	$T(y)$	One planned statistical inference
	2. Test pre-chosen from set of possible tests	$T(y; \phi)$	One test with pre-registered ϕ
	3. Test based on the data	$T(y; \phi(y))$	Only one test. Different test would have been performed given different data
	4. Fishing	$T(y; \phi_j)$	Performing j tests and reporting the best one(s)

ϕ : control variables, covariates, transformations, data coding rules, exclusion, outliers, main effects, interactions, subgroups, alternate outcomes, direction of effect

TOP 3 FORKING PATHS

- One reasonable hypothesis maps on to several reasonable statistical hypotheses (i.e., one to many)
 - PONV is associated with age
 - Statistical interaction (moderation) must be taken into consideration for the primary interpretation
 - Age x sex is needed to consider the effect of age
 - Adding confounder control post hoc
 - We should control for several covariates as they seem to be confounding the age association
-

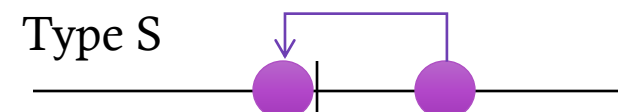
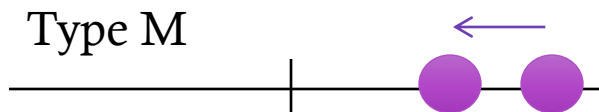
DEFINITION OF FALSE

Traditional

- *Type I error*: Reject null hypothesis when it is true
- *Type II error*: Fail to reject null hypothesis when it is false

Gelman

- Type M error: Errors in the magnitude of the estimated effect size
- Type S error: Errors in the sign of the estimated effect size



RECIPE FOR GREATER CHANCES OF BEING FALSE

- Small effect size
 - Small (modest) sample size
 - Large measurement error
 - High variation
 - Low prior probability of effect
-

RECOMMENDATIONS:

- Avoid forking paths
 - Pre-registration
 - Fixed statistical analysis plans
 - Primary designation
 - Moderators
 - Multiplicity adjustments
- Reproducible documents

REPORTING AND INTERPRETATION OF RANDOMIZED CONTROLLED TRIALS WITH STATISTICALLY NONSIGNIFICANT RESULTS FOR PRIMARY OUTCOMES

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

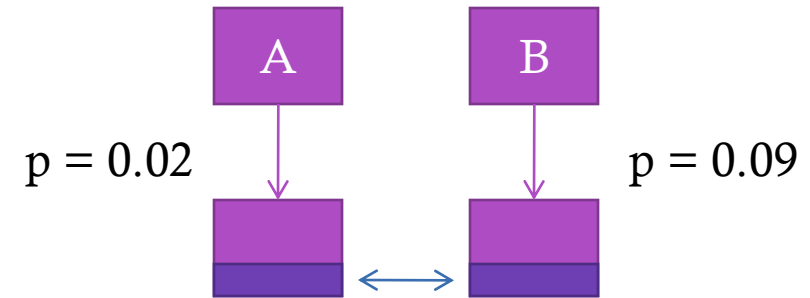
- Frequency and degree of “Spin” in published RCTs
- Spin[†]: specific reporting that could distort the interpretation of results and mislead readers.
 - From ignorance of scientific (statistical) issue
 - Unconscious bias
 - Willful intent to deceive

[†]Fletcher RH, Black B. “Spin” in scientific writing: scientific mischief and legal jeopardy. *Med Law*. 2007; 26(3):511-525.

REPORTING AND INTERPRETATION OF RANDOMIZED CONTROLLED TRIALS WITH STATISTICALLY NONSIGNIFICANT RESULTS FOR PRIMARY OUTCOMES

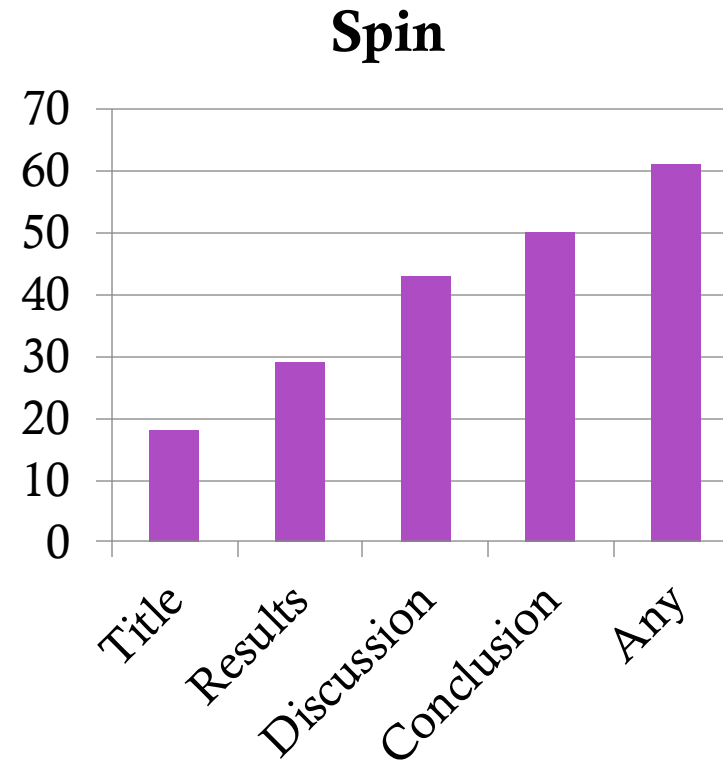
Strategies of Spin

- A focus on statistically significant results
 - within-group comparison
 - secondary outcomes
 - subgroup analyses
 - modified population of analyses
- Interpreting statistically nonsignificant results for the primary outcomes as showing treatment equivalence or comparable effectiveness
- Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results.



REPORTING AND INTERPRETATION OF RANDOMIZED CONTROLLED TRIALS WITH STATISTICALLY NONSIGNIFICANT RESULTS FOR PRIMARY OUTCOMES

- 1735 RCTs identified from December 2006
 - N = 72 Included in analysis
 - All were two group comparisons with nonsignificant primary findings
- 61.1% of manuscripts had spin in at least one section



Background—Patent foramen ovale (PFO) is prevalent in patients with migraine with aura. Observational studies show that PFO closure resulted in migraine cessation or improvement in ~80% of such patients. We investigated the effects of PFO closure for migraine in a randomized, double-blind, sham-controlled trial.

Methods and Results—Patients who suffered from migraine with aura, experienced frequent migraine attacks, had previously failed ≥ 2 classes of prophylactic treatments, and had moderate or large right-to-left shunts consistent with the presence of a PFO were randomized to transcatheter PFO closure with the STARFlex implant or to a sham procedure. Patients were followed up for 6 months. The primary efficacy end point was cessation of migraine headache 91 to 180 days after the procedure. In total, 163 of 432 patients (38%) had right-to-left shunts consistent with a moderate or large PFO. One hundred forty-seven patients were randomized. No significant difference was observed in the primary end point of migraine headache cessation between implant and sham groups (3 of 74 versus 3 of 73, respectively; $P=0.51$). Secondary end points also were not achieved. On exploratory analysis, excluding 2 outliers, the implant group demonstrated a greater reduction in total migraine headache days ($P=0.027$). As expected, the implant arm experienced more procedural serious adverse events. All events were transient.

Conclusions—This trial confirmed the high prevalence of right-to-left shunts in patients with migraine with aura. Although no significant effect was found for primary or secondary end points, the exploratory analysis supports further investigation. The robust design of this study has served as the model for larger trials that are currently underway in the United States and Europe. (*Circulation*. 2008;117:1397-1404.)

DISCUSSION

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